Embedded Control for Robotics

Lecture # 10-11-12 Discrete time control System

Book: Embedded Digital Control with Microcontrollers

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Overview Today's Lecture

- Inverse Z-transform
- Relations between s-plane and z-plane
- Solution of difference Equations

The inverse Z-transform

- The inverse Z-transform is the mathematical operation that converts a Z-transform expression from the complex frequency domain (Z-domain) back into its original discrete time-domain signal, denoted as (x[n]).

- It is used to find the time-domain representation of a signal, often for a system's impulse response,

- It is computed using methods
  1. partial fraction expansion,
  2. long division,
  3. or using a table of known transform pairs.

Inverse Z-transform

1. **Long Division:** We first use long division to obtain as many terms as desired of the z-transform expansion.

2. **Partial Fraction:** This method is almost identical to that used in inverting Laplace transforms. However, because most z-functions have the term z in their numerator, it is often convenient to expand $F(z)/z$ rather than $F(z)$.

Long Division Inverse Z-transform

- **Example-3:** Obtain the inverse z-transform of the function

$$F(z) = \frac{z+1}{z^2 + 0.2z + 0.1}$$

- **Solution**

$$z^2 + 0.2z + 0.1 \overline{)\, z + 1 \;\Big|\; z^{-1} + 0.8z^{-2} - 0.26z^{-3} + \ldots\ldots}$$

$$z + 0.2 + 0.1z^{-1}$$

$$0.8 - 0.10z^{-1}$$

$$0.8 + 0.16z^{-1} + 0.08z^{-2}$$

$$-0.26z^{-1} - \ldots$$

Continue…

$$F(z) = \frac{z+1}{z^2 + 0.2z + 0.1}$$

- *Inverse z-transform*

$$F(z) = 0 + z^{-1} + 0.8z^{-2} - 0.26z^{-3} + \cdots$$

$$f(k) = \{0, \quad 1, \quad 0.8, \quad -0.26, \quad \ldots \}$$

Partial Fractions Inverse Z-transform

- **Example-4:** Obtain the inverse z-transform of the function

$$F(z) = \frac{z+1}{z^2 + 0.3z + 0.02}$$

- **Solution**

$$\frac{F(z)}{z} = \frac{z+1}{z(z^2 + 0.3z + 0.02)}$$

$$\frac{F(z)}{z} = \frac{z+1}{z(z^2 + 0.1z + 0.2z + 0.02)}$$

Inverse Z-transform

$$\frac{F(z)}{z} = \frac{z+1}{z(z+0.1)(z+0.2)}$$

$$\frac{F(z)}{z} = \frac{A}{z} + \frac{B}{z+0.1} + \frac{C}{z+0.2}$$

$$A = z\frac{F(z)}{z}\bigg|_{z=0} = F(0) = \frac{1}{0.1\times 0.2} = \frac{1}{0.02} = 50$$

$$B = (z+0.1)\frac{F(z)}{z}\bigg|_{z=-0.1} = (z+0.1)\frac{1}{z}\frac{z+1}{(z+0.1)(z+0.2)}\bigg|_{z=-0.1} = \frac{-0.1+1}{(-0.1)(-0.1+0.2)} = -90$$

$$C = (z+0.2)\frac{F(z)}{z}\bigg|_{z=-0.2} = (z+0.2)\frac{1}{z}\frac{z+1}{(z+0.1)(z+0.2)}\bigg|_{z=-0.2} = \frac{-0.2+1}{(-0.2)(-0.2+0.1)} = 40$$

Inverse Z-transform

$$\frac{F(z)}{z} = \frac{50}{z} - \frac{90}{z+0.1} + \frac{40}{z+0.2}$$

$$F(z) = 50 - \frac{90z}{z+0.1} + \frac{40z}{z+0.2}$$

- Taking inverse z-transform (using z-transform table)

$$f(k) = 50\delta(k) - 90(-0.1)^k + 40(-0.2)^k$$

Example 5: LTI System Analysis in z Domain

- Given the following system function:

$$H(z) = \frac{1+0.25z^{-1}}{1+0.8z^{-1}-0.84z^{-2}}$$

(a) Plot the pole-zero diagram of $H(z)$.

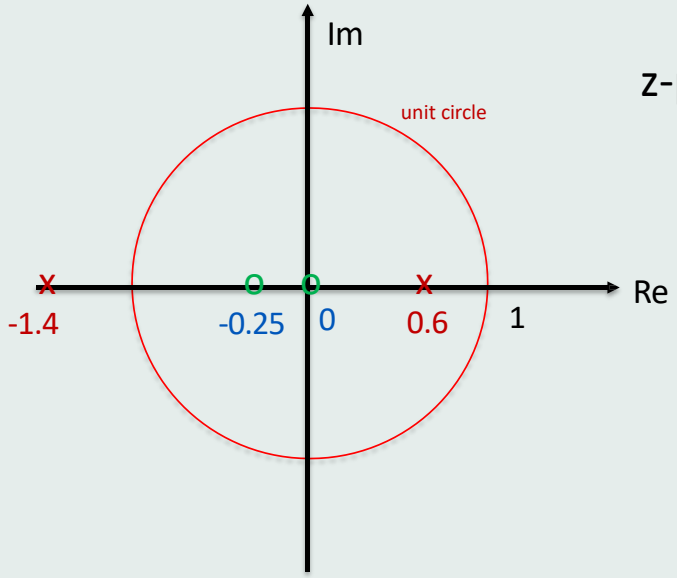
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Solution (a)

Plot the **pole-zero diagram** of $H(z)$.

$$H(z) = \frac{1 + 0.25z^{-1}}{1 + 0.8z^{-1} - 0.84z^{-2}} = \frac{1 + 0.25z^{-1}}{(1 + 1.4z^{-1})(1 - 0.6z^{-1})} \frac{z^2}{z^2} = \frac{z(z + 0.25)}{(z + 1.4)(z - 0.6)}$$

- Two zeros at z = 0 and z = -0.25
- Two poles at z = -1.4 and z = 0.6



Home Work

- For each of the following equations, determine the order of the equation and then test it for (i) linearity, (ii) time invariance, (iii) homogeneity.

a) $y(k+2) = y(k+1)y(k) + u(k)$

b) $y(k+3) + 2y(k) = 0$

c) $y(k+4) + y(k-1) = u(k)$

d) $y(k+5) = y(k+4) + u(k+1) - u(k)$

e) $y(k+2) = y(k)u(k)$

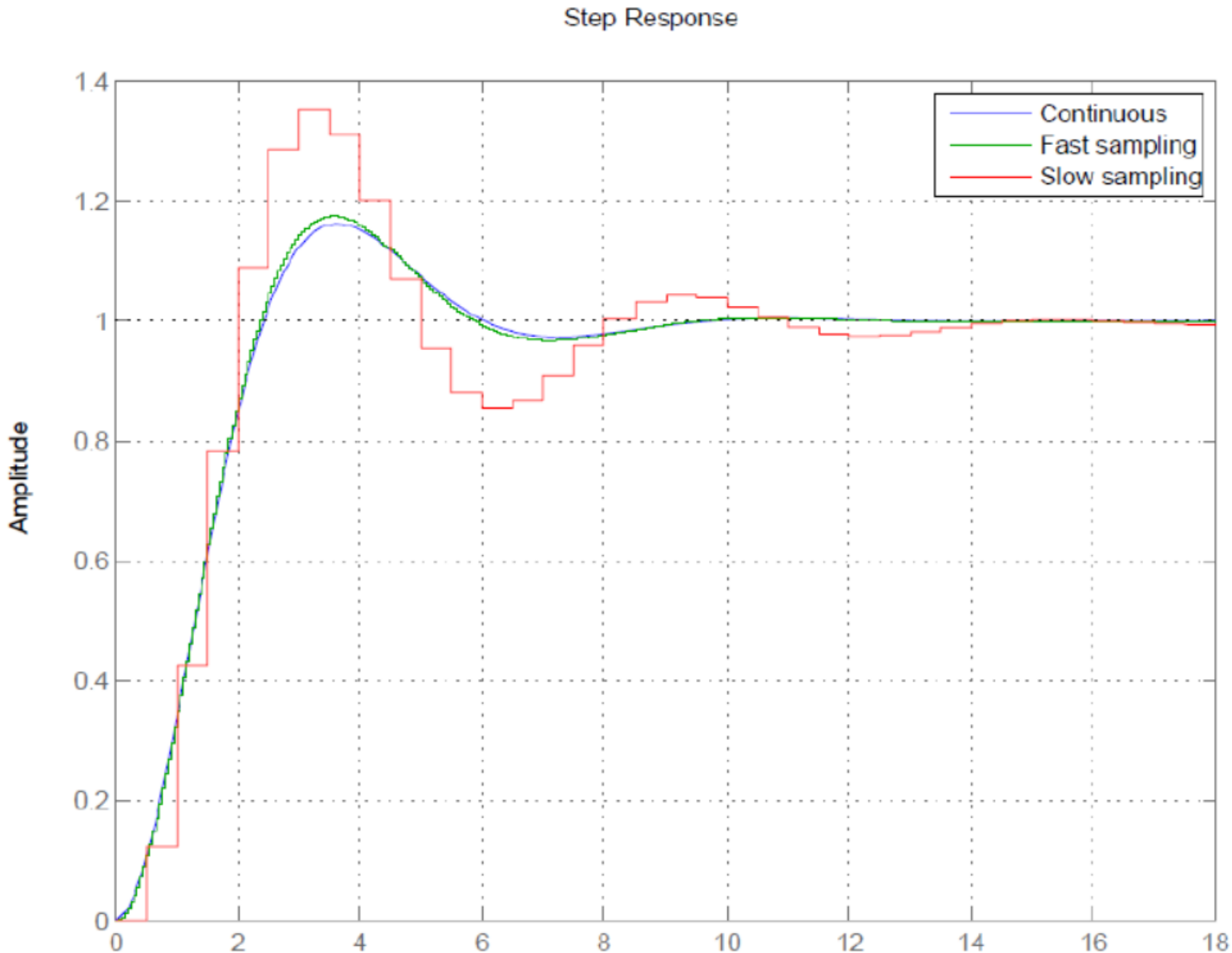
Home Work

- Find the z-transforms of the following sequences

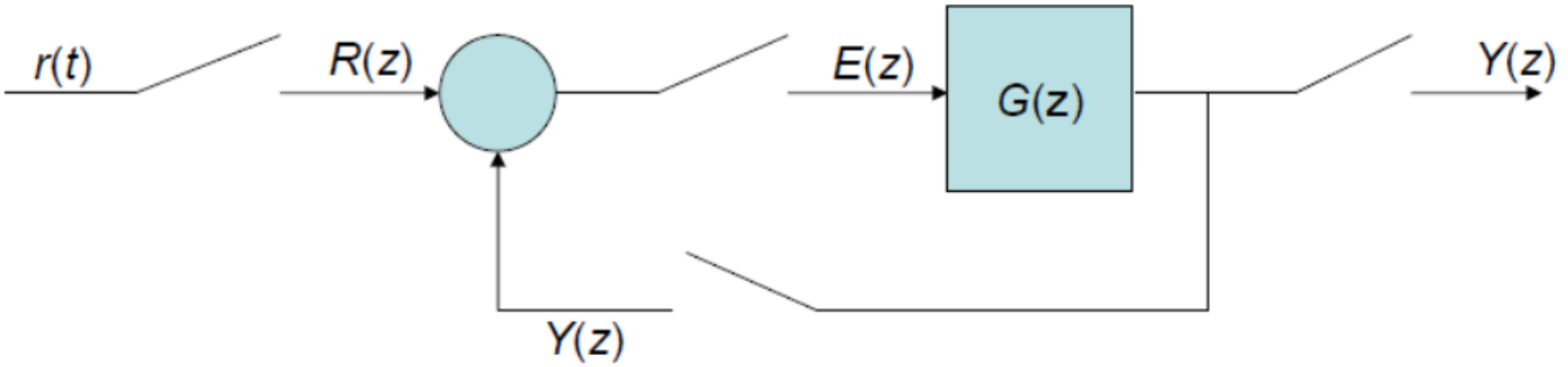
a) $\{0, 1, 2, 4, 0, 0, \dots\}$

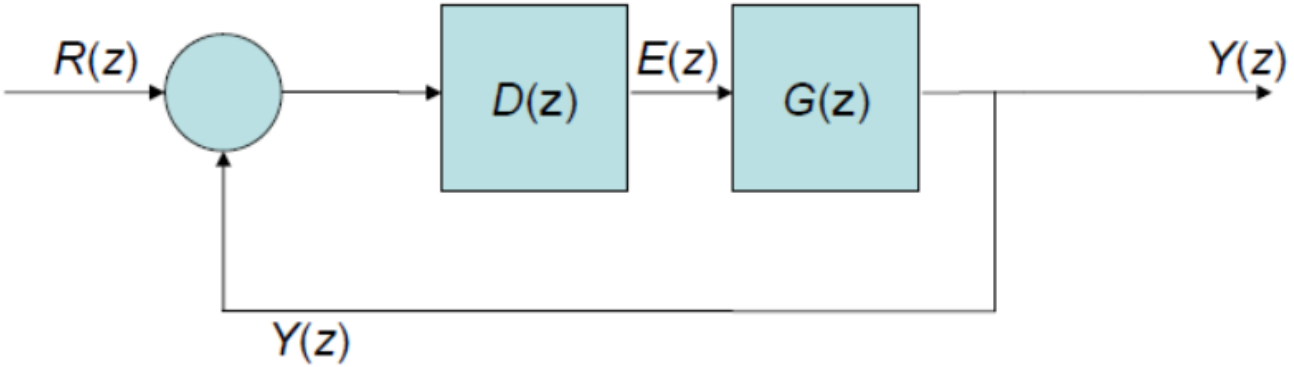
b) $\{0, 0, 0, 1, 1, 1, 0, 0, 0, \dots\}$

c) $\{0, 2^{-0.5}, 1, 2^{-0.5}, 0, 0, 0, \dots\}$


Step Response
Amplitude
Continuous
Fast sampling
Slow sampling
0
0.2
0.4
0.6
0.8
1
1.2
1.4
0
2
4
6
8
10
12
14
16
18

Closed-Loop Feedback Sampled-Data Systems





$$\frac{Y(z)}{R(z)} = T(z) = \frac{G(z)}{1+G(z)} = \frac{G(z)D(z)}{1+G(z)D(z)}$$

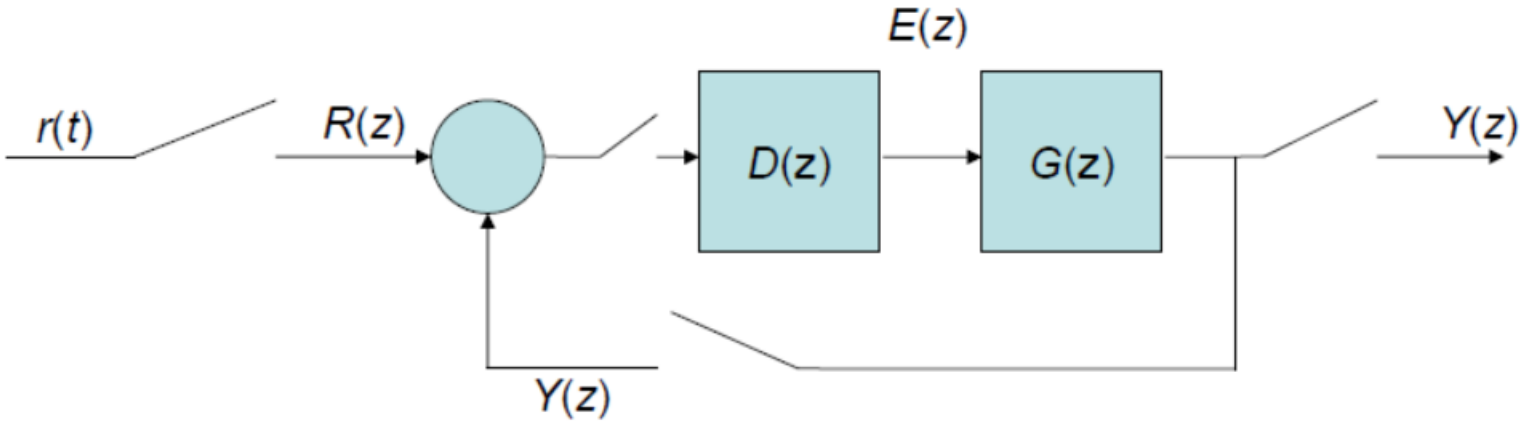
Design Of Digital Control System

Steps

- Start with continuous system.
- Add sampled-data system elements.
- Chose sample period, usually small but not too small. Use sampling period T = 1 / 10 f_B, *where* $f_B = \omega_B / 2\pi$ and ω_B is the bandwidth of the closed-loop system.
  - Practical limit for sampling frequency: $20 < \omega_s / \omega_B > 40$
- Digitize control law.
- Check performance using discrete model or SIMULINK.

Start with a Continuous Design

$D(s)$ may be given as an existing design or by using root locus or bode design.



Add Samples Necessary for Digital Control

- Transform $D(s)$ to $D(z)$: We will obtain a discrete system with a similar behavior to the continuous one.
- Include D/A converter, usually a zero-order-device.
- Include A/D converter modeled as an ideal sampler.
- And an antialiasing filter, a low pass filter, unity gain filter with a sharp cutoff frequency.
- Chose a sample frequency based on the closed-loop bandwidth ω_B of the continuous system.

Closed-Loop System with Digital Computer Compensation

$$\frac{Y(z)}{R(z)} = T(z) = \frac{G(z)D(z)}{1+G(z)D(z)}$$

The tranfer function of the computer is $\frac{U(z)}{E(z)} = D(z)$

Consider the second order systemwith a zero-order hold and a plant

$Gp(s) = \frac{1}{s(s+1)}$ when $T=1; G(z) = \frac{0.3678(z+0.7189)}{(z-1)(z-0.3678)}$; If we select $D(z) = \frac{k(z-0.3678)}{(z+r)}$

We cancer the pole of $G(z)$at $z = 0.3678$ and have the two parameters r and K.

$$D(z) = \frac{1.359(z-0.7189)}{(z+0.240)}; G(z)D(z) = \frac{0.5(z+0.7189)}{(z-1)(z+0.240)}$$

$$G_c(s) = K\frac{s+a}{s+b};\ D(z) = C\frac{z-A}{z-B}; Z\{G_c(s)\} = D(z); A = e^{-aT}; B = e^{-bT};\ C\frac{(1-A)}{(1-B)} = K\frac{a}{b}$$